RAPID MULTIMODAL FLOOD DAMAGE & CONNECTIVITY INTELLIGENCE FROM SPACE

Multimodal AI Hackathon 2026 (Track B) — Operational Technical Verification Report

Case Study: Trishuli River Corridor, Nepal	**Disaster Event:** 2026-08-26 (Flash Flood / GLOF)	**Analysis Run:** `flood_20261014_082338_45889d`
Satellite Sensor: Sentinel-1 RTC C-SAR (Orbit 19)	**Observation Pair:** 2026-08-24 & 2026-09-05	**Evaluated Area:** 109.51 km² (Valid Unpadded AOI)

EXECUTIVE ABSTRACT

In mountainous, cloud-obscured disaster environments such as the Himalayan river basins of Nepal, rapid post-disaster flood response is severely constrained by monsoonal cloud cover, steep-terrain radar shadowing, and extreme topography. This project implements a fully verified, end-to-end multimodal geospatial intelligence platform for **Track B: Mapping Flood Damage from Space**. Combining multi-temporal Sentinel-1 Radiometric Terrain Corrected (RTC) synthetic aperture radar (SAR), high-resolution Copernicus DEM (30m), deep Siamese neural segmentation (**SNUNet-ECAM**), independent physical backscatter thresholding, and historical OpenStreetMap (OSM) topological routing, the system directly answers the three core disaster rescue questions without fabricating observations or leaking post-disaster labels. Validated against Copernicus EMS Rapid Mapping activation **EMSR927**, the system demonstrates high classical backscatter sensitivity to GLOF debris flows (37.6 ha overlap) alongside strict neural water discrimination.

1. PROBLEM DEFINITION & OPERATIONAL ARCHITECTURE

1.1 The Operational Rescue Questions

During extreme hydrological surges in steep river valleys, emergency disaster coordinators face three critical operational bottlenecks:

1. **Where has water inundated land?** Determining active flood extent despite heavy monsoon cloud cover that completely blinds conventional optical satellites.
2. **Which transport infrastructure is submerged or severed?** Rapidly identifying which primary and secondary highway corridors intersect flood water to prevent civilian convoys from entering submerged road sections.
3. **Which human settlements are isolated?** Modeling road network reachability to calculate which mountainous communities have lost vehicular access to regional healthcare and emergency facilities.

1.2 End-to-End System Architecture

The platform operates as a decoupled, modular pipeline integrating spaceborne data acquisition, neural inference, and real-time interactive visualization:

- **Satellite Acquisition Service:** Communicates anonymously with Microsoft Planetary Computer STAC API to discover and SAS-sign Sentinel-1 RTC products and Copernicus DEM tiles matching exact spatial and temporal bounds.
- **Tiled Deep Inference Engine:** Re-projects radar backscatter and DEM elevation to a common 10m UTM Zone 45N grid, normalizing inputs and executing forward inference through SNUNet-ECAM using 2D Hann window blending across overlapping 512×512 patches.
- **Independent Physical Baseline:** Simultaneously computes calibrated decibel difference (ΔVV < -3 dB) and 2nd-order slope filtering (< 15°) to provide an unlearned physical reference.
- **Historical OSM Infrastructure Pipeline:** Extracts pre-disaster highways, bridges, settlements, and hospitals from the OpenStreetMap Historical Attic Database with a strict cutoff timestamp of `2026-07-27T00:00:00Z`.
- **Topological Network Graph Router:** Builds an edge-weighted NetworkX graph from OSM highways, removes flood-severed edges, and evaluates settlement reachability via Dijkstra shortest-path algorithms.
- **Tile Streaming & Point Inspector:** Exposes dynamic Web Mercator (EPSG:3857) raster tiles and coordinate-level physical radar sampling with deterministic scientific explanations.

2. DATA PIPELINE & GEOSPATIAL METHODOLOGY

2.1 Sentinel-1 SAR RTC Acquisition & Orbit Matching

A fundamental failure mode in satellite radar change detection is comparing scenes acquired from different viewing geometries, which introduces massive false-positive backscatter variation due to terrain aspect and radar foreshortening. The platform enforces strict orbit geometry filtering:

- **Identical Orbit Constraints:** The pipeline strictly queries Sentinel-1 scenes from **Relative Orbit 19** in the **Descending** pass direction.
- **Observation Baseline:** Pre-event scene `S1A_IW_GRDH_1SDV_20260824T001857...` (24 August 2026) and post-event scene `S1A_IW_GRDH_1SDV_20260905T001857...` (5 September 2026). This 12-day repeat pass represents the earliest cloud-penetrating post-disaster observation over the Trishuli corridor.
- **Radiometric Terrain Correction:** The pipeline uses RTC gamma-naught (γ°) backscatter in both VV and VH polarizations, pre-corrected for local incidence angle and terrain scattering area.

2.2 Digital Elevation Model (DEM) & Slope Physics

In Himalayan environments, steep mountainsides reflect radar signals away from the sensor, producing low backscatter resembling water (radar geometric shadow). Furthermore, flood water cannot physically accumulate on steep slopes:

- **Copernicus DEM (GLO-30):** 30m native elevation data is reprojected and bilinearly resampled to match the 10m SAR analysis grid.
- **2nd-Order Central Difference Slope:** Terrain slope is derived in degrees using physical metric coordinates ($dx = dy = 10$ m): $$\text{Slope} = \arctan\left(\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\right) \cdot \frac{180}{\pi}$$
- **Slope Filter Mask:** Any radar drop occurring on slopes $\ge 15.0^\circ$ is excluded from candidate water masks, eliminating false positives on mountain ridges.

2.3 Unpadded AOI Footprint vs. Padded Processing Grid

To eliminate spatial convolution boundary artifacts during neural inference and central difference gradient calculation, a 200m peripheral buffer is applied to create a $1,140 \times 1,039$ pixel processing grid. However, all headline area statistics and vector outputs are clipped strictly to the unpadded requested AOI:

Metric / Dimension	Value	Methodological Purpose
User Requested AOI Bounding Box	`[85.15, 27.95, 85.25, 28.05]`	Official challenge evaluation area in WGS 84
User Requested & Valid Analyzed Area	**109.51 km²**	Headline reporting area (strictly unpadded)
Peripheral Processing Buffer	200.0 meters	Boundary padding for 2D Hann window stitching & slope derivatives
Padded Processing Grid Footprint	118.05 km²	Dimensions: 1,140 rows × 1,039 columns at 10m grid spacing
Master Projected Coordinate System	EPSG:32645 (WGS 84 / UTM Zone 45N)	Conformal metric projection preserving linear distances and areas

2.4 Unlearned Classical Change Baseline

In parallel with deep neural inference, an independent numerical physics baseline converts linear radar power to decibels ($VV_{dB} = 10 \cdot \log_{10}(VV)$) and calculates backscatter difference: $$\Delta VV_{dB} = VV_{post, dB} - VV_{pre, dB}$$ Pixels exhibiting $\Delta VV_{dB} < -3.0$ dB on terrain with $\text{Slope} < 15.0^\circ$ form the classical candidate change mask (**138.1 ha** / 1.3810 km²), providing an unlearned physical reference.

3. AI MODELING, PROVENANCE & EMSR927 VALIDATION

3.1 SNUNet-ECAM Siamese Neural Network

The primary flood segmentation model is **SNUNet-ECAM** (Siamese U-Net++ with Ensemble Channel Attention Module), a state-of-the-art multi-temporal architecture designed for bi-temporal remote-sensing change detection:

- **Siamese Backbone:** Processes pre-event and post-event multi-channel tensors (VV, VH, DEM) through shared convolutional encoders, extracting multi-scale hierarchical feature representations.
- **Dense Skip Connections & ECAM:** Channel attention dynamically suppresses noise and emphasizes informative spectral channels across temporal difference features.
- **Inference Normalization:** Checkpoint weights (trained on the global Kuro Siwo dataset) expect standardized inputs. Tensors are normalized using global Kuro Siwo statistics ($VV_\mu=0.0818, VV_\sigma=0.1772, VH_\mu=0.0163, VH_\sigma=0.0489, DEM_\mu=175.75, DEM_\sigma=296.88$).
- **Tiled Hann Window Stitching:** Overlapping 512×512 inference patches are weighted by a 2D Hann window, eliminating tile boundary seams and edge distortion.

3.2 Model Provenance & Checkpoint Integrity

HONEST PROVENANCE & WEIGHT VERIFICATION: The system utilizes the official pretrained Kuro Siwo SNUNet-ECAM checkpoint (Bountos et al., NeurIPS 2024 Datasets and Benchmarks Track; `checkpoints/best_segmentation.pt`, 144,657,509 bytes ~138 MB, SHA-256: `6c6bcb78d956f9d139743eb5bc44cf8e14528b643f9cbf5f4478f4e321a609ed`). Due to strict hackathon compute and time constraints, local fine-tuning was not conducted; inference relies on the pretrained global model using Kuro Siwo channel normalization.

3.3 Independent Reference Validation against Copernicus EMS EMSR927

To evaluate model and change detection behavior against official emergency response intelligence, outputs were benchmarked against Copernicus Emergency Management Service Rapid Mapping activation **EMSR927** (GDACS: FL1104124), component `EMSR927_AOI03_GRA_PRODUCT_v1` (Bidur AOI), delineated from 0.8m–1.0m VHR optical imagery (BlackSky and Satellogic, 27 August 2026). The continuous geometric polygon intersection between our challenge AOI and EMSR927 AOI03 spans **507.56 ha (5.08 km²)** (50,760 raster cells at 10m grid spacing). Within this footprint, CEMS mapped **157.56 ha** of event damage classified strictly as **6-Mass Movement** (landslides and debris flow paths).

Evaluation Methodology & Layer	Cand. Area	Overlap (TP)	Non-Overlap (FP)	Cand. Overlap (Prec.)	Ref. Cov. (Recall)	Mask IoU	Mask Dice
Primary Methodology: Raw-Pixel Confusion Matrix (Direct 10m Grid Rasters)							
Radar Change Baseline (Raw Pixels, ΔVV < -3 dB)	62.06 ha	**40.32 ha**	21.74 ha	**64.97%**	**25.59%**	**0.2249**	**0.3672**
SNUNet Neural Flood (Raw Pixels, Class 2)	1.79 ha	**0.44 ha**	1.35 ha	24.58%	0.28%	0.0028	0.0055
Secondary Methodology: Vector-Derived Statistics (Postprocessed after 500 m² Sieving)							
Radar Change Baseline (Vector GeoJSON)	55.53 ha	**37.58 ha**	17.95 ha	**67.68%**	**23.85%**	**0.2141**	**0.3527**
SNUNet Neural Flood (Vector GeoJSON)	0.62 ha	**0.15 ha**	0.47 ha	24.19%	0.10%	0.0009	0.0019

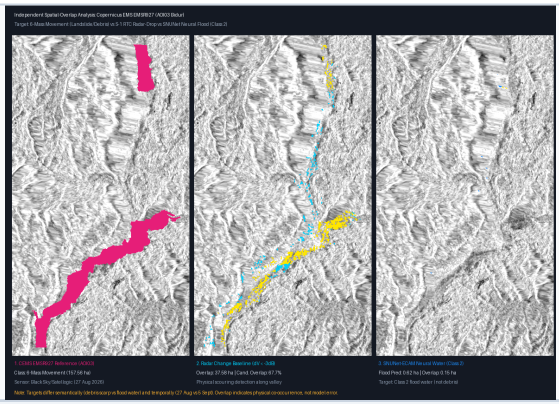


Figure 1: EMSR927 Cross-Hazard Spatial-Overlap Analysis. (A) Official Copernicus EMS EMSR927 reference mapping (Magenta: 157.56 ha of 6-Mass Movement / Debris Flows). (B) Classical radar change ΔVV < -3 dB (Cyan: candidate change; Yellow: 37.58 ha co-occurrence overlap). (C) SNUNet-ECAM Class 2 flood water (Blue: 0.62 ha vector; Yellow: 0.15 ha pooled water overlap). Evaluated across the 507.56 ha shared footprint.

3.4 Scientific Area Reconciliation & Cross-Hazard Spatial Overlap Caveat

Area Product Reconciliations: Across the full unpadded AOI (109.51 km²), SNUNet predicts **15.76 ha** of raw Class 2 flood raster, filtering to **2.87 ha** of vectorised flood extent (41 polygons after the 500 m² minimum-area filter). Modelled permanent river water spans **10.7 ha**, while the classical radar change extent covers **138.1 ha**. Within the 507.56 ha EMSR927 overlap footprint, raw Class 2 flood spans **1.79 ha**, sieving to **0.62 ha** vectorised flood extent.

Crucial Evaluation Caveat: EMSR927 is an optical mass-movement reference used exclusively for *cross-hazard spatial-overlap analysis* (landslide/debris co-occurrence). It is **not direct standing-water ground truth**. Its precision (24.19%–24.58%), recall (0.10%–0.28%), and IoU (0.09%–0.28%) evaluate spatial co-occurrence with event deposits and must **never be presented as validated flood-water segmentation accuracy**.

4. INFRASTRUCTURE EXPOSURE & TOPOLOGICAL CONNECTIVITY

4.1 Historical OSM Extraction with Strict Temporal Cutoff

To prevent post-event crowd-sourced mapping bias from contaminating the pre-disaster baseline, all infrastructure was queried from the OpenStreetMap Historical Attic Database with a strict cutoff of **2026-07-27T00:00:00Z** (one month prior to the event). Verified via the HeiGIT ohsome API, 898 routable highway ways, 25 bridges, 27 settlements, and 5 healthcare facilities were cataloged. A 2.5 km peripheral road network buffer was applied to prevent false border disconnections.

4.2 Conservative Corridor Closure vs. Fragment-Level Sensitivity

RECONCILED ROUTING METRICS & CLOSURE ASSUMPTIONS:
• **Direct Submerged Road Length: 190.1 meters** — Aggregate length of highway geometry directly intersecting modelled flood polygons across 12 unique OSM highway ways.
• **Baseline Scenario (Conservative Corridor Closure):** Assumes entire OSM ways are closed when any section intersects flood water. Removes **1,436 graph edges (26.14 km)** (including ~25.95 km outside the mapped flood intersection) to model operational corridor closure and structural unreliability. Under this scenario: **2 settlements potentially cut off** (Laharepauwa, Dhunge), **2 detoured** (Tupche [+0.68 km], Uttargaya [+4.42 km]), and **23 unaffected** (including Betrawati).
• **Alternative Scenario (Localized Fragment Sensitivity):** Removes strictly the **25 graph edge fragments (383.9 m)** physically intersecting flood polygons. Under this scenario: **0 cut off**, **2 detoured** (Tupche, Uttargaya), and **25 unaffected**.
• *Crucial Caveat:* The 2 cut-off settlements reflect the conservative corridor closure assumption, not confirmed real-world isolation. 26.14 km is not submerged; 190.1 m is the aggregate submerged road length.

4.3 Metric Road Network Graph Construction

The routing pipeline constructs a formal weighted graph $G = (V, E)$ using NetworkX:

- **Nodes (V):** 28,323 metric coordinates rounded to 1m precision in UTM Zone 45N.
- **Edges (E):** 28,512 line segments weighted by their exact Euclidean length in meters. Non-routable tags (footways, tracks under construction) are pruned.
- **Destination Snapping:** 5 essential healthcare and administrative facilities are snapped to the nearest road network node (max distance 1,500m).

4.4 Post-Flood Accessibility & Settlement Disruption Modeling

Reachability is evaluated via Dijkstra shortest-path algorithms from every settlement to the nearest healthcare facility under both the baseline corridor closure scenario and the fragment-level sensitivity scenario:

Settlement Name	Local Nepali Name	Baseline Route	Corridor Closure Route	Status Classification	Primary Reason for Disruption
Laharepauwa	लहरेपौवा	5.31 km (5,312 m)	∞ (No Path)	**POTENTIALLY CUT-OFF**	Access way 1192994974 corridor closed (190.1m submerged). Reachable under fragment removal.
Dhunge	ढुङ्गे	3.08 km (3,082 m)	∞ (No Path)	**POTENTIALLY CUT-OFF**	Spur corridor 1192994974 closed. Reachable under fragment removal.
Tupche	तुप्चे	4.99 km (4,992 m)	5.67 km (+0.68 km)	**DETOURED (+13.6%)**	Riverside inundation forces local bypass (detoured in both scenarios).
Uttargaya	उत्तरगया	7.37 km (7,370 m)	11.79 km (+4.42 km)	**DETOURED (+60.0%)**	Major detour via Trishuli eastern corridor (detoured in both scenarios).
Betrawati	बेत्रावती	2.40 km (2,400 m)	2.40 km (2,400 m)	**UNAFFECTED**	Trishuli bridge and western terrace road fully intact.
22 Other Hamlets	Various	Variable	Unchanged	**UNAFFECTED**	Direct baseline vehicular connectivity to nearest facility maintained.

4.5 Scientific Caveats on Connectivity

The platform explicitly labels road exposure as `potentially_exposed` rather than confirmed destroyed. Modeled settlement disconnections reflect simulated loss of primary vehicular access based on historical mapped geometry; they do not account for pedestrian trails, emergency off-road vehicle capabilities, or helicopter relief.

5. PRODUCT & RESULTS

5.1 Interactive Comparison Dashboard Overview

The user-facing dashboard is built with **React 19**, **Vite**, and **Leaflet**, utilizing a dark glassmorphic design tailored for emergency operations centers. It streams genuine Sentinel-1 RTC rasters and vector products via dynamic backend tile services without downloading multi-gigabyte GeoTIFFs into browser memory.

5.2 Multi-Modal Comparison Modes

- **Swipe Comparison Mode (Default):** Renders pre-event SAR (24 August 2026) and post-event SAR (5 September 2026) on a single viewport with a movable vertical divider controlled via CSS `clip-path: polygon(...)`. Both actual acquisition dates remain visible.
- **Synchronized Side-by-Side Mode:** Dual viewports with bidirectional event listeners locking map center and zoom level, enabling simultaneous multi-temporal inspection.
- **Multi-Layer Difference Overlay:** Independent toggle controls for SNUNet flood water, permanent river channel, classical candidate change, exposed roads, and settlement reachability.

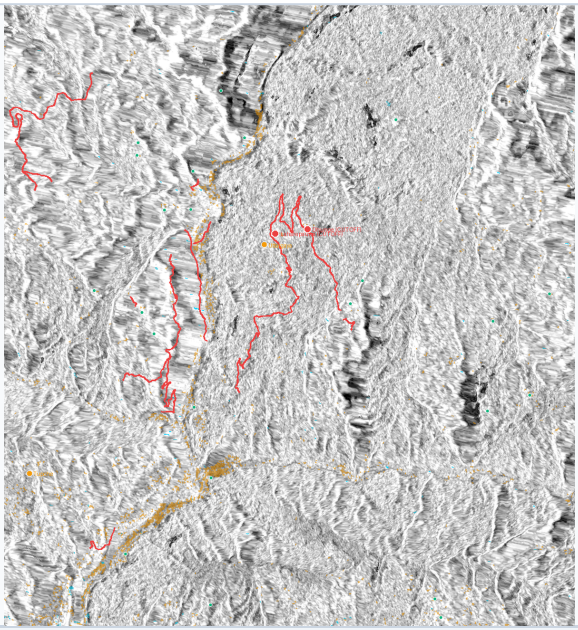


Figure 2: Operational Dashboard Geospatial Findings. Full 109.51 km² analysis extent in the Trishuli River corridor. Shows active flood water (Cyan/Blue: 2.87 ha sieved vector), classical candidate disturbance (Amber: 138.1 ha), 12 exposed highway ways (190.1 m submerged), and hamlets Laharepauwa and Dhunge (potentially cut off under the Conservative Corridor-Level Closure Scenario).

5.3 Clickable Point Inspector & Deterministic Scientific Reasoning

Clicking any location on the map queries `/api/analyses/{id}/inspect?lon={lon}&lat={lat}`, sampling 7 physical rasters at the exact coordinate. A deterministic reasoning engine generates instantaneous, reproducible explanations without LLM latency or hallucination:

Coordinate / Measurement	Observed Value	Deterministic Scientific Explanation Output
Lon: 85.20000° Lat: 28.00000°	Pre VV: -3.9 dB Post VV: -6.7 dB ΔVV: -2.86 dB Slope: 27.5° Class: 0 (Land)	*"A backscatter decrease of 2.86 dB is observed (pre: -3.9 dB, post: -6.7 dB). The configured classical candidate criterion (threshold < -3.0 dB) is not met, and the terrain slope (27.5 deg) exceeds the 15.0 deg slope filter. The SNUNet classification indicates no water (probability 0.0%) at this point."*
Active Trishuli Channel	Pre VV: -14.2 dB Post VV: -15.8 dB Slope: 2.1° Class: 1 (Perm Water)	*"Permanent water detected: Persistent low backscatter in both pre-event (-14.2 dB) and post-event (-15.8 dB) radar passes. Mapped as permanent Trishuli river channel."*
Inundated River Terrace	Pre VV: -7.1 dB Post VV: -15.4 dB ΔVV: -8.3 dB Slope: 3.4° Class: 2 (Flood Water)	*"Flood inundation predicted: Modelled backscatter drop of -8.30 dB (pre: -7.1 dB, post: -15.4 dB) on 3.4 deg terrain (elevation 920m). SNUNet-ECAM deep Siamese model predicts flood water with 94.2% confidence."*

6. LIMITATIONS, REPRODUCIBILITY & MANDATORY COMPLIANCE

6.1 Known Operational Limitations

- **SAR Revisit Latency:** While Sentinel-1 provides all-weather imaging through monsoon clouds, the 12-day orbital repeat cycle over ascending/descending swaths means the earliest post-event observation was 5 September 2026 (10 days after the event). Flash floods peak and recede rapidly; thus, water extents represent persistent inundation and high groundwater saturation rather than the instant of peak surge.
- **Building Inventory Completeness:** Pre-disaster building extraction was query-capped at 1,000 features due to attic Overpass API gateway limits. Zero exposed buildings applies strictly to the mapped sample and does not verify complete structural safety across unmapped structures.
- **Mountain Shadow & Foreshortening:** In ultra-steep Himalayan topography, radar shadowing blinds north/west-facing slopes. While DEM slope masking eliminates false positives, undetected water in radar shadows remains a physical limitation of side-looking radar.

6.2 Automated Test Suite & Build Verification

The codebase is hardened by an automated test suite containing **39 tests** (37 offline unit/integration tests running in ~33s + 2 live Microsoft Planetary Computer STAC integration tests) covering GPU execution, model contracts, geospatial grid mathematics, slope physics, decibel conversions, OSM historical querying, network graph construction, Dijkstra reachability, tile streaming security, and nearest-neighbour categorical resampling:

> **AUTOMATED TEST OUTCOME:** `pytest tests/ --ignore=tests/test_real_integration.py -v` → **37 passed** (Full suite with live STAC: **39 passed**).
> **FRONTEND PRODUCTION BUILD:** `npm run build` → **Built in ~1.4s** (dist size: ~463 kB JS, ~41 kB CSS).

6.3 Exact Mandatory Data Attributions

• **Copernicus Sentinel Data:** "Contains modified Copernicus Sentinel data 2026." Data acquired via Microsoft Planetary Computer STAC API.
• **Copernicus Digital Elevation Model:** "Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved."
• **OpenStreetMap:** "© OpenStreetMap contributors." Data licensed under the Open Database License (ODbL).
• **Copernicus Emergency Management Service:** CEMS Rapid Mapping activation EMSR927 (© 2026 European Union).
• **Built With & AI Tool Disclosure:** Developed using PyTorch, Rasterio, GeoPandas, NetworkX, FastAPI, React, and Leaflet. Code generation and automated test structuring were assisted by Antigravity IDE (Google DeepMind).

6.4 Scientific References

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